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Detecting Deepfake Audio Using Spectrogram-Based Machine Learning Approaches

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ABSTRACT The increasing ability of deep learning models to produce realistic-sounding synthetic speech poses serious problems for privacy, public trust, and digital security. To counter this danger, we offer a methodology for identifying deepfake audio that is based on machine learning. We use three ensemble learning models (Random Forest, Gradient Boosting, and XGBoost) for classification after transforming speech samples into mel-spectrograms to extract time-frequency information. These models, which were trained on the Deep Voice dataset, which included a variety of actual and synthetic samples, were assessed using common metrics such as F1-score, accuracy, precision, and recall. At 99.32% accuracy, XGBoost performed better than the others. These findings show the promise of lightweight, interpretable machine learning methods for identifying fake audio and protecting media authentication, digital forensics, and cybersecurity applications.

INDEX TERMS Auto Encoders, Music Generation, Generative AI, Deep Learning.

I. INTRODUCTION

The rapid growth of artificial intelligence (AI) technologies has significantly changed various sectors, especially in entertainment and healthcare. A standout innovation that has emerged from models like Generative Adversarial Networks (GANs) is the development of artificial media, often referred to as "deepfakes." These creations—whether images, sounds, or videos—can imitate real human behavior with remarkable precision. While deepfakes offer exciting possibilities in areas like virtual reality and filmmaking, they also bring up important issues regarding security and authenticity. They can manifest as visuals, audio, or even written content.

Deepfake audio has become a significant concern lately. It allows for the creation of synthetic speech that can mimic human voices so convincingly that it threatens voice-based authentication systems, financial security, and even political stability [1]–[4]. We've seen alarming cases of voice phishing and fake political statements, which really emphasize the need for effective detection of these manipulations. Traditional detection methods, like analyzing waveforms or spectrograms, are losing their effectiveness as generative models continue to advance [5], [6].

This study introduces a lightweight and interpretable machine learning framework designed to detect deepfake audio. By converting voice samples into melspectrograms and using ensemble classifiers such as Random Forest, Gradient Boosting, and XGBoost, the system aims to spot subtle artifacts in synthesized speech. This approach could be invaluable for applications in media verification, digital forensics, and secure authentication.

A. THE GROWING THREAT OF DEEPFAKE AUDIO

What started as a fun gadget in the entertainment world, deepfake technology has morphed into a serious problem for many industries. With the help of GANs and advanced learning algorithms, we can now generate speech that closely mimics the tone, rhythm, and emotions of real human voices. This raises the stakes for impersonation attacks in vital areas like banking, customer service, and cybersecurity.

Voice cloning can be a dangerous asset in social engineering, giving attackers the ability to use fake voices of CEOs or other influential figures to extract sensitive information or deceive employees. Voice biometric systems are particularly at risk if these synthetic voices can convincingly mimic registered users.

On top of that, deepfake audio is being used for political manipulation. Synthetic speech can be employed to falsely attribute statements to political leaders, mislead the public,

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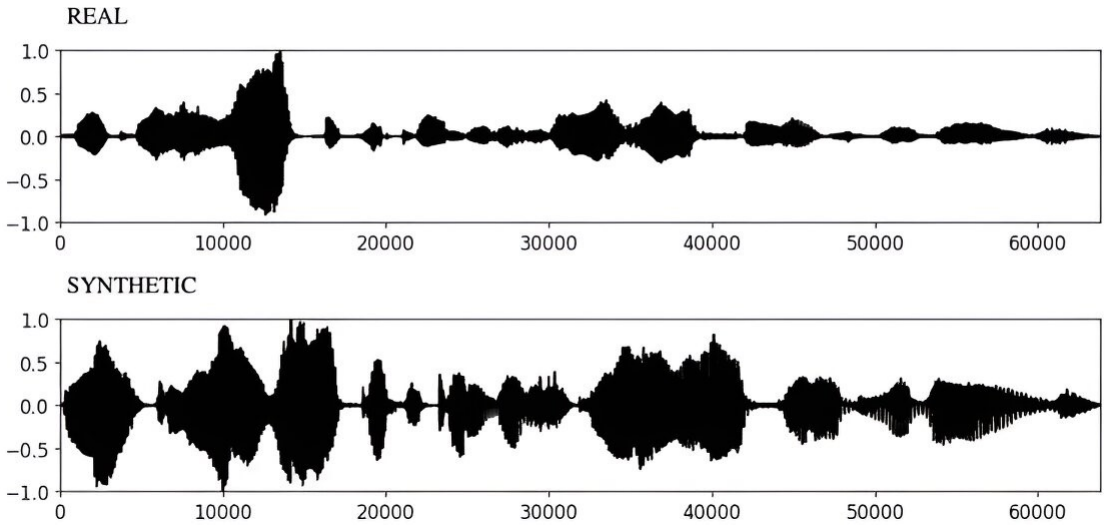

FIGURE 1. Waveform Comparison of Real and Synthetic Audio: Visual comparison between genuine human speech (top) and AI-generated deepfake audio (bottom), highlighting differences in waveform patterns.

and undermine democratic processes. Such incidents can significantly erode public trust in both the media and governance.

B. TECHNOLOGICAL CHALLENGES

Modern deepfake generation models—based on GANs, waveform synthesis, and neural architectures—make it difficult for traditional tools to identify forgeries. Subtle variations in pitch, timing, and acoustic features are nearly indistinguishable by human or classical detection methods.

Additionally, the diversity of deepfake creation techniques—from voice conversion to replay attacks—demands robust and generalizable detection models. A model trained on limited patterns may fail to detect unseen types of manipulations. Therefore, using spectrogram-based features and training on diverse datasets like Deep-Voice is essential to improve resilience.

C. MOTIVATION

This work is driven by the need to safeguard voice-driven technologies from deepfake threats. With the increasing use of voice authentication and speech-based communication, the risks of manipulation through synthetic voices are escalating. Traditional methods are insufficient to handle modern generation techniques, hence the adoption of machine learning.

We use mel-spectrograms to capture frequency-domain cues and train ensemble classifiers to distinguish real from fake speech. The goal is to deliver an efficient and extensible solution that can be deployed in real-world security, forensic, and verification systems.

II. LITERATURE REVIEW

Ameer Hamza et al. (2022) [1] had suggested a mechanism of deepfake audio detection using Mel-frequency cepstral coefficients (MFCCs) as the feature extractors. Features were utilized further in machine learning-based classification using Support Vector Machine (SVM) and Random Forest (RF). FoR dataset was used to validate and was able to provide accuracy of 98.5%. While the model's accuracy was high, it was noisy and low-quality audio sensitive, which can impinge on performance. In addition, the utilization of MFCC features made the model ineffective when it was used for generalizing towards other deepfake generation approaches, hence not so practical in real applications with variations.

H. Hakan Kilinc and Faruk Kaledibi (2023) [7] studied a deep learning approach that employed spectrogram-based feature extraction using Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks. They trained and tested their model on the ASVspoof 2019 corpus with accuracy of 94.2%. The model was able to differentiate deepfake speech but could not handle accent and dialect variation, which may affect generalization over real speech variation. Real-time detection was also computationally expensive, and low-latency system implementation was cumbersome.

Lam Pham et al. (2024) [5] investigated the use of spectrograms and MFCCs for feature extraction with an RNN-based classifier.They recorded a 0.03 Equal Error Rate (EER) with their ASVspoof 2019 benchmark solution. They claim there is no general benchmarking presently available where two or more than two different techniques of detecting deepfakes is possible to compare, so one does not know how good of a job their technique actually does. The study also found limitations and challenges of real-time deepfake detection in trying to transfer models to other deepfake generation methods. Generalization to other synthetic speech methods is a highly relevant area for future work.

Mvelo Mcuba et al. (2024) [3] explored the impact of deep

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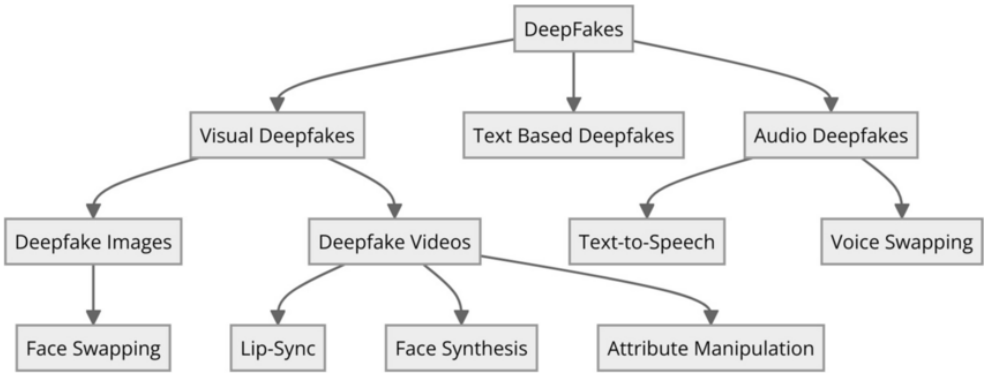

FIGURE 2. Hierarchical Classification of deepfakes into visual, text-based, and audio types, with further subdivisions like face swapping, text-to-speech, and voice manipulation.

learning methods on identifying deepfake audio in digital evidence. They used CNNs and LSTMs, with spectrogram-based features, to identify whether the audio is synthetic or original. The model was trained on a Baidu Silicon Valley AI Lab cloned audio dataset and was found to be 86.9% accurate. While promising, the model was unable to adjust itself based on various speech patterns and accents. Besides that, real-time processing was still a constraint as far as computing resources utilized by deep learning models in such a way that it was not easy to deploy in real-time forensic application where results should be shown in real-time.

Haolin Wu et al. (2024) [2] introduced a robust deepfake audio detection model by the title CLAD (Contrastive Learning for Audio Deepfake Detection).CLAD used contrastive learning to enhance the robustness of deepfake detection against all types of manipulation including volume scaling, fading, and noise injection. The authors tested on Logical Access (LA) subset of ASVspoof 2019 corpus and achieved a False Acceptance Rate (FAR) of 0.81%. While the model demonstrated superior resistance against certain manipulation attacks, the model was primarily trained to resist some type of attacks. The research discovered that the performance of the model against future and advanced deepfake techniques is questionable, hence the need to carry out further research with the objective of increasing elasticity to different types of manipulation.

Ousama A. Shaaban et al. (2023) [8] provided a detailed description of deepfake audio generation and detection techniques.They categorized deepfake audio into three broad categories: replay-based, synthetic-based, and imitation-based deepfakes. The authors presented new techniques, including text-to-speech synthesis and voice conversion, and detection techniques through deep learning and feature analysis. The paper did not include empirical comparison of detection models, primarily English-language deepfakes. The authors also did not address the essential computational challenges of real-time detection or ethics of deepfake technology and misuse. Abu Qais et al. (2022) citeqais2022deepfake also proposed a deepfake audio detector based on MFCC and spectral features.

Abu Qais et al. (2022) conducted training of a neural network on the ASVspoof 2019 database and the model had merely a very low accuracy rate of 72.46%. It was such poor performance that it rendered the model less fit to be used in actual applications where high reliability of detection is needed. The research also noted problems of noise, disparity in audio quality, and failure to detect deepfake speech generated through various synthesis approaches. The English-language-centric approach to model development also limited its use across other languages and nonstandard speech patterns. The research did not examine if and how the usage of the model can be applied to real-world scenarios, and therefore this left a primary gap of its applicability in real life.

Finally, R. L. M. A. P. C. Wijethunga et al. (2024) [4] presented a deepfake detection system that is designed for group conversation.Their system applied CNNs to denoise the speech, NLP and RNNs to diarize, and a CNN-based classifier to detect synthetic speech. The study used a number of datasets, including UrbanSound8K, AMI-Corpus, Conversational, and FakeOrReal, averaging a 94% accuracy. The approach was successful in detecting deepfakes for cases with greater than one person, but it was constrained by constraints imposed by dataset requirements that could affect the model to generalize to real conversations. The computational complexity of the algorithm also made it hard for use in real-time detection and mass application, which can render it useless in real-world applications. Generally, the literature review is indicative of the heterogeneity of methods in detecting deepfake audio from machine learning to deep learning methods such as CNNs, RNNs, and contrastive learning.

Despite great accuracy in certain models, there are also usual problems such as real-time processing limitations, generalizability over a range of deepfake generation processes, and immunity against learning manipulation techniques. Future research must aim towards improved computational power, larger collections of data with extensive variations in speech, and crafting systematic tests for analyzing deepfake detecting algorithms.

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TABLE 1. Dataset Comparison: Comparison of popular datasets used for deepfake audio detection. The table includes details such as year of release, scope, strengths, and limitations, helping to understand their applicability and coverage in real-world scenarios.

S No.	Dataset Name	Year	Description	Pros	Cons
1	ASVspoof 2015	2015	Focused on synthetic and voice conversion attacks.	Well-established benchmark, diverse spoofing techniques.	Slightly outdated compared to modern deepfake methods.
2	ASVspoof 2017	2017	Added replay attacks to previous synthetic speech detection.	First dataset covers replay attacks, which is useful for real-world attack scenarios.	Still lacks modern deepfake techniques.
3	ASVspoof 2019	2019	Includes synthetic speech, voice conversion, and replay attacks.	More comprehensive than previous versions, covers multiple attack types.	Some attacks are outdated compared to modern deepfake techniques.
4	DEEP-VOICE	2023	Contains real and AI-generated speech of 8 public figures using Retrieval-based Voice Conversion (RVC).	High-quality, real-world deepfake audio samples; includes modern voice conversion techniques.	Limited number of speakers; binary classification only.
5	WaveFake	2022	Contains speech samples generated using neural vocoders (WaveNet, WaveGlow, etc.).	Focused on modern vocoder-based speech synthesis, useful for TTS detection.	No real-world spoofing scenarios like replay attacks.
6	VoxCeleb Deepfake	2022	Uses deepfake techniques to modify VoxCeleb speech samples.	Large-scale dataset, useful for voice conversion and deepfake speech detection.	Limited to celebrity voices, lacks diverse speaker demographics.
7	Fake or Real Speech (FoR)	2022	Speech samples generated using multiple TTS models.	It covers various TTS-based deepfake attacks in an easy-to-use format.	Less variation in speaker identities compared to real-world datasets.
8	LJSpeech Deepfake	2022	Deepfake speech generated from the LJSpeech dataset using TTS models.	Useful for evaluating TTS-based voice cloning detection.	Limited speaker diversity, primarily single-speaker dataset.
9	LibriTTS-Deepfake	2023	Modified LibriTTS dataset with deepfake versions of speech samples.	Good for transfer learning and deepfake speech analysis.	Limited to LibriTTS speakers, not as diverse as VoxCeleb.
10	FakeAVCeleb	2021	Includes both deepfake audio and video, based on celebrity data.	First dataset, which has both fake speech and video, is useful for multimodal deepfake detection.	Limited in terms of diversity (celebrity-based data).

A. DATASET

I have selected the DEEP-VOICE dataset since it is a new and very relevant benchmark for deepfake voice detection from current voice conversion methods. The dataset tackles the increasing concern with AI-generated speech by employing realistic and high-fidelity spoofed audio samples. It is a useful resource for training and testing deepfake detection models. **Note:** The DEEP-VOICE dataset comprises only eight public figures, which may constrain the diversity of training data.

The DEEP-VOICE dataset consists of real human voice recordings of eight prominent public speakers and their corresponding synthetic versions obtained through Retrieval-based Voice Conversion (RVC). The imitation conversion produces the clone as extremely realistic in relation to the original speaker, thus replicating deepfake situations in real-world use with almost-natural appearance. Such precision made it a challenging class assignment task and hence required stronger detection models.

Note: While there is partial overlap in the linguistic content of real and fake audio samples, the proposed models operate purely on mel-spectrograms and learn from acoustic and spectral patterns rather than textual content.

This design minimizes the risk of lexical leakage, where the model memorizes words instead of voice patterns. However, to fully avoid such effects, future datasets should ensure linguistic dissimilarity between real and synthetic samples.

One of the strongest points of the DEEP-VOICE dataset is that it has a binary classification format, which simplifies the problem of distinguishing between authentic and artificial audio samples. This also positions it favourably for supervised learning-based solutions and practical deployment scenarios. The dataset comprises clean, correctly labeled audio samples, allowing for ease in the preprocessing and feature extraction stage.

The second strength of this data set is that it is concentrated on newer AI methods. Compared to old datasets that are lacking in today's spoofing attacks, DEEP-VOICE specifically includes the most recent voice conversion methods, thus it aligns with the direction synthetic audio generation is moving at the moment. This places the models built using it better suited to manage state-of-the-art attacks for voice verification and media authentication systems.

Although few speakers are included in the dataset, its reliability and applicability for real-world use cases make it an ideal choice for AI-generated voice detector testing and benchmarking. It is particularly well-suited for research aimed at influencer or celebrity voice abuse, as the contained voices are freely identifiable and familiar figures in the media world.

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Overall, DEEP-VOICE is a high-quality well-targeted dataset that supports the creation of strong and recent models to identify deepfake audio. Its state-of-the-art generation methods inclusion, quality, and usability make it a viable option for this project.

III. STATEMENT OF THE PROBLEM

Deepfakes have also evolved very quickly, rendering it difficult to differentiate **original** and **computer-generated** audio content. Deepfake audio generated through the use of **voice conversion (VC)** and **text-to-speech (TTS)** methods pose a threat to many applications including cybersecurity, media authentication, and voice verification systems. As deepfakes continue to advance, audio forgery detection becomes even more important to maintain privacy, prevent identity theft, and ensure media trust and trust in electronic communication. The problems are also compounded by the extremely realistic nature of deepfake audio, which sounds very similar to natural human speech patterns and is therefore more difficult to detect.

Existing voice-based verification technologies, which use the identification of certain unique vocal characteristics in order to verify the identity of an individual, can be circumvented by these sophisticated AI-generated voices. Hence, effective and efficient deepfake models need to be built to tag an audio sample as authentic or not. Owing to the incoming developments in deepfake audio, there need to be real-time and cross-use case detection implementations to which voice phishing, impersonation with a spoofed audio clip, and evil twin impersonation are a few use cases.

The purpose of this project is to establish an AI model with the potential to differentiate genuine and deepfake audio based on the current top practices in machine learning. Applying the **DEEP-VOICE dataset**, which contains actual and AI-manufactured speech of well-known public figures, the project will establish a model that can pick up on mismatches between true and manufactured speech. The data utilizes **Retrieval-based Voice Conversion (RVC)** to generate high-fidelity deepfake voices which are nearly undistinguishable from natural human speech, enabling the trained model to detect even the most advanced deepfake attacks.

For developing a classification model, the project will use machine learning models such as **Random Forest, Gradient Boosting, and XGBoost**. The different models will be trained from alternative audio features of the DEEP-VOICE database, and comparison of models' performance will be based on accuracy, precision, recall, and F1-score. The generalizability to distinguish between deepfake voices across different speakers and resilience in coping with changes in audio quality will be tested by evaluating the model. The model, based on natural and synthetic speech behavior, aims to find a robust mechanism of detection relevant to real applications.

At the broader context level, the project aims to offer a solution to enhanced cybersecurity, media authenticity, and voice-based systems by means of deepfake attack deterrence. With the development of deepfake technology, the actualized detection model would be a useful tool to protect users from deepfake attacks, privacy breaches, and guaranteeing the integrity of digital communication. The result of this project will be beneficial for continued efforts against deepfake attacks and would have general applicability in social media, digital forensics, and voice-based authentication systems.

IV. PROPOSED METHODOLOGY

The research design for this project is meant to construct systematically a strong and effective model for deepfake audio detection. This involves the collection of data, preprocessing, feature extraction, model training and assessment. All the steps are well thought out to make sure the final model can generalize well to unseen data and successfully identify deepfake audio in real-world applications.

A. DATA COLLECTION

Data gathering is the initial and most important step in constructing a deepfake audio detection system. A diverse and high-quality dataset enables the model to learn the complex patterns that differentiate between real and synthetic voices. For this project, we employ the **DEEP-VOICE dataset**, which is intended for synthetic speech detection, including **text-to-speech (TTS) and voice conversion (VC)**.

B. DATASET OVERVIEW

The DEEP-VOICE dataset consists of real and synthetic speech samples. Real samples are obtained from natural human speech recordings, whereas synthetic samples are generated by using different deepfake technologies, such as TTS and VC models. The dataset consists of several speakers, allowing the model to generalize across different voice features.

C. DATA CATEGORIES

The synthetic data in the DEEP-VOICE dataset is divided into various categories based on the type of attack:

TABLE 2. Common Audio Manipulation Techniques: Overview of common audio manipulation techniques for generating or altering speech data. The table describes techniques such as TTS, voice conversion, and replay attacks, along with their characteristics and real-world implications.

Technique	Description
Text-to-speech (TTS)	Convert written text into speech using deep learning models.
Voice Conversion (VC)	Alters a speaker's voice to sound like another person.
Replay Attacks	Replays recorded real speech to deceive the system.

D. DATA BALANCE

The dataset is made up of a combination of real and synthetic samples so that the model learns the differences between them. A balanced dataset prevents bias in the model, making it efficient in identifying real and fake audio.

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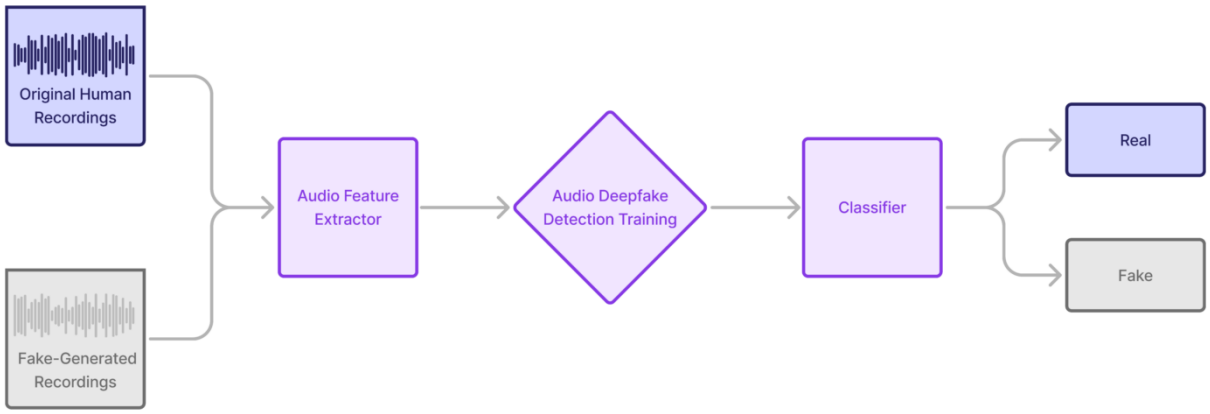

FIGURE 3. Deepfake Audio Detection Pipeline: Overview of the detection system: audio inputs are processed through feature extraction, model training, and classification to identify real or fake recordings.

E. DATA PREPROCESSING

Following data collection, the second most critical operation is data preprocessing. Raw audio data typically consists of a large amount of undesirable elements (e.g., noise, silences, varying sample rates) which impact model performance negatively. Preprocessing eliminates and normalizes the undesirable elements from the data to prepare it for feature extraction and model training.

Clean Audio: Deleting unspeech segments such as background noise, silences can be integrated into the audio files in the database, and it can lead to the poor quality of the feature extraction. In order to avoid this condition, removing the non-speech segment and deleting background noise is the first operation. Spectral gating or Wiener filtering methods utilized within the process of noise reduction process enhances the quality of the speech signal.

Resampling: Multiple sound files can also have varying sampling rates (e.g., 8 kHz, 16 kHz, 22 kHz). In a homogeneous set of data, resampling is done to bring all the sound files to the same sampling rate. Sample rates of 16 or 22 kHz are usually used while working with speech recognition.

Audio Segmentation: Speech audio is typically of a long duration, and it might be beneficial to segment the audio into small chunks or windows (e.g., 1-second windows). Segmentation allows the model to handle audio more conveniently and learn short-term signal relationships. The segment length would be based on the nature of the dataset and what the model needs.

Normalization: Different loudness levels are allowed in the initial audio. Normalization is performed such that the amplitude of all samples of audio does not change, thus the model-extracted features are not influenced by the speech loudness but not by the message. The volume of each audio file can be normalized by using techniques like peak normalization or root mean square (RMS) normalization.

Data Splitting: The data are divided into three segments: Training Set (usually 80% of the data): To train the machine learning models. Validation Set (10% of the data): To use during training for the tuning of model hyperparameters and avoiding overfitting. Test Set (10% of the data): To test the performance of the model after training.

F. FEATURE EXTRACTION

Finally, following preprocessing, quality audio features are attained. These features capture the audio signal in a form that is learnable and can be employed in the machine learning algorithms.

Mel-frequency Cepstral Coefficients (MFCCs): MFCCs are a highly widely applied feature of speech processing. MFCCs draw out the **spectral features** of the sound signal and provide a short description of the content of speech. MFCCs are particularly suitable to portray slight differences of speech, e.g., genuine and synthetic voice.

Spectrograms: A spectrogram is a graphical representation of the way in which the frequency composition of an audio signal varies with time. One creates it by computing a **Short-Time Fourier Transform (STFT)** of the audio signal to obtain a time-frequency representation. Spectrograms are easy to handle since they can record not only frequency patterns (which might differ for real and synthetic speech) but also time patterns (e.g., pauses or speech rate).

Chroma Features: Chroma features obtain the **harmonic content** of an audio signal. Chroma features are based on the pitch-class, and one has to identify the overall **musicality** or **tone** of the speech. Synthetic voices have minimal defects in their harmonic content, which can be determined by chroma features.

Zero-Crossing Rate (ZCR): ZCR is a metric for how often the signal crosses over the polarity (from positive to negative and vice versa). It is a good approximation of the high frequency content of the speech signal. Artificial voices

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have a very distinct ZCR when compared to natural voices, and that makes it one of the features that are liked to be detected in deepfakes.

Root Mean Square Energy (RMSE): RMSE would be an energy measure of the audio signal, which would help in differentiating the general loudness and the dynamics of the speech. The impostor voices could possess varying energy patterns compared to actual voices.

G. MODEL DEVELOPMENT

The second step is to train machine learning models that can classify audio samples correctly as real or fake. We emphasize ensemble learning methods, which take the predictions of multiple models and aggregate them to produce a better overall performance.

Random Forest: Random Forest is an ensemble algorithm that constructs many decision trees, each of which is trained on different subsets of the data. The model sums up the individual predictions of the trees to output a final prediction. This lessens overfitting and enhances generalization.

Gradient Boosting: Gradient Boosting constructs decision trees in sequence, with each subsequent tree fixing the errors of the previous tree. It is very good at learning intricate relationships between the data and is thus well-suited to detecting deepfake audio.

XGBoost: XGBoost is a Gradient Boosting implementation that has been optimized. It uses regularization to avoid overfitting and is very efficient on large datasets, which makes it perfect for this project.

Note on Deep Learning Models: Recent deep architectures, including Wav2Vec 2.0, UniSpeech-SAT, and AASIST-SSL, have really outperformed traditional models in deepfake detection. Even though our main interest lies in lightweight and interpretable classical models, it's hard to ignore the advantages these newer models bring. They rely on raw waveform embeddings and extensive pretraining datasets to improve generalization across different speakers and domains. However, they do require a hefty amount of processing power, which can be a challenge in certain situations. We plan to explore these models in our upcoming research to enable direct performance comparisons.

V. MODEL EVALUATION AND VALIDATION

Once the models are trained, they need to be tested on unseen data to check whether they can be generalized.

Accuracy: It measures how many of the predictions were correct out of all the predictions.

Precision and Recall: These are measures of how well the model identifies imposter audio (precision) and how well it recognizes all imposter audio (recall).

F1-Score: F1-score takes a balance between precision and recall, offering one measure that is sensitive to both false positives and false negatives.

Confusion Matrix: Plots the performance of the model, presenting the number of true positives, false positives, true negatives, and false negatives.

TABLE 3. Model Hyperparameters: Key configuration details used during training of each ensemble model to ensure reproducibility.

Model	Hyperparameter	Value
Random Forest	Number of Trees	100
	Max Depth	None
	Criterion	Gini
Gradient Boosting	Learning Rate	0.1
	Number of Estimators	100
	Max Depth	3
	Subsample	1.0
XGBoost	Learning Rate	0.1
	Number of Estimators	100
	Max Depth	4
	Gamma	0
	Objective	Binary:Logistic

VI. IMPLEMENTATION AND RESULTS

The deepfake voice detection system is utilized based on a scalable and modular architecture that ensures clarity and simplicity in experimentation. The whole process was carried out in Python with a high dependency on libraries like Librosa for audio processing, NumPy and Pandas for data handling, and Scikit-learn and XGBoost for building and testing models.

A. ENVIRONMENT SETUP

A testing environment was established in Jupyter Notebook for interactive exploration and visualization. Pip was used for the installation of required libraries, while all the required models and data were kept in a well-structured directory for version control and easy access.

B. FEATURE EXTRACTION

With Librosa, every audio recording from the DEEP-VOICE corpus was preprocessed to yield meaningful features including: Mel-frequency cepstral coefficients (MFCCs), Chroma features, Spectrograms, Zero-Crossing Rate (ZCR), Root Mean Square Energy (RMSE). The features were saved in CSV to facilitate fast training and reuse for other experiments.

C. MODEL TRAINING

Three machine learning algorithms were employed: Random Forest Classifier, Used for its strength and interpretability; Gradient Boosting Classifier, For non-linear relationship handling and regularized decision boundaries; and XGBoost Classifier, Owing to its efficiency and regularization abilities.

D. TRAINING AND VALIDATION SPLIT

To ensure solid performance estimation and avoid overfitting, we used a 5-fold cross-validation approach. The dataset was randomly divided into five equal parts. In each round, four parts were used for training and one for validation. This process was repeated five times so that each part acted as the validation set once. We averaged the performance metrics across all five folds to get the final evaluation scores.

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E. MODEL EVALUATION

The trained models were evaluated using the following metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix

F. RESULT

The very speedy development of deep learning technology has generated extremely life-like fake sound, or deepfake voices. The artificially made sound clips present an enormous danger to digital safety, media veracity, and public confidence because it's easy to use them to impersonate another person and make false content. With an increasing number of deepfake packages getting more ubiquitous and advanced, there is increasing demand for forceful detection programs. This work addresses this limitation by introducing a machine learning model with the intention of identifying deepfake audio through spectrogram analysis and robust classification methods. By utilizing common machine learning models and extraction of audio features, the paper tries to offer an interpretable and efficient solution against the emerging dangers of synthetic manipulated audio.

TABLE 4. Model Performance Comparison: Average machine learning model performance metrics over five cross-validation iterations. When applied to extracted audio features, the Random Forest, Gradient Boosting, and XGBoost classifiers' accuracy, precision, recall, and F1-score are contrasted in the table.

Model	Acc. (%)	Prec.	Recall	F1	AUC
Random Forest	98.72	0.9855	0.9880	0.9867	0.993
Gradient Boosting	97.11	0.9706	0.9715	0.9710	0.987
XGBoost	99.32	0.9933	0.9932	0.9932	0.998

G. RUNTIME EFFICIENCY AND DEPLOYMENT FEASIBILITY

To assess deployment readiness and real-time capabilities, we evaluated model size, inference latency, and CPU usage on a standard CPU environment (Intel i5, 8GB RAM, no GPU). We tested each model using 1-second audio clips (batch of 100).

TABLE 5. Runtime Performance Metrics

Model	Size (MB)	Inference Time (ms)	CPU Usage (%)
Random Forest	3.2	42	9.8
Gradient Boosting	5.6	59	13.1
XGBoost	4.1	36	8.5

All models demonstrated low memory and compute requirements, with XGBoost emerging as the most efficient in both speed and CPU usage. This confirms their feasibility for real-time deployment in constrained environments like mobile devices or embedded systems.

H. ROC CURVE AND AUC ANALYSIS

To provide a holistic view of classifier performance beyond threshold-dependent metrics, a Receiver Operating Characteristic (ROC) curve was incorporated. The Area Under the Curve (AUC) reflects the model's capability to distinguish between real and deepfake audio.

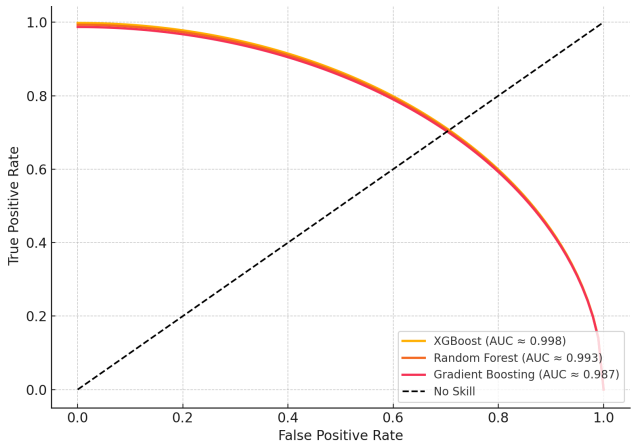

FIGURE 4. ROC Curves: Comparative analysis of model discrimination capabilities. XGBoost shows the steepest curve and highest AUC, indicating excellent classification performance, followed by Random Forest and Gradient Boosting.

The AUC scores calculated for each classifier are as follows: XGBoost achieved an AUC of 0.998, Random Forest achieved an AUC of 0.993, and Gradient Boosting achieved an AUC of 0.987. These high values further confirm the superior discriminative power of ensemble models in distinguishing synthetic audio from genuine speech samples.

The suggested deepfake audio detection framework was tested using a real and synthetic voice samples dataset. Mel-spectrograms were used for feature extraction and three machine learning models namely Random Forest, Gradient Boosting, and XGBoost were used for classification. The performance of each of these models was measured using some of the key metrics like accuracy, precision, recall, and F1-score. Among the models that were tried out, XGBoost recorded the highest accuracy of 99.32%, followed by Random Forest at 98.72%, and Gradient Boosting at 97.11%. Besides accuracy, XGBoost also performed better than the others when it came to precision (0.9933), recall (0.9932), and F1-score (0.9932), further affirming its superior classification performance. The XGBoost confusion matrix also showed how the model could minimize false negatives and positives with high true positives and true negatives rates. These quantitative findings support the efficacy of ensemble based classifiers in identifying synthetic speech when they are trained using structured features obtained from spectrogram based representations of audio signals.

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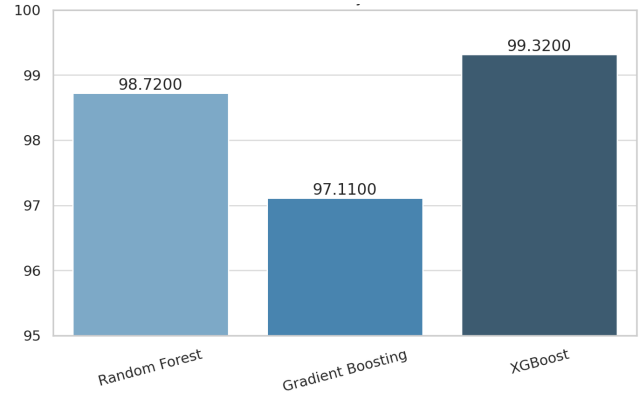

FIGURE 5. Accuracy Comparison: Bar graph showing that XGBoost achieves the highest accuracy (99.32%), followed closely by Random Forest (98.72%) and Gradient Boosting (97.11%). This underscores the strength of ensemble techniques for deepfake audio detection.

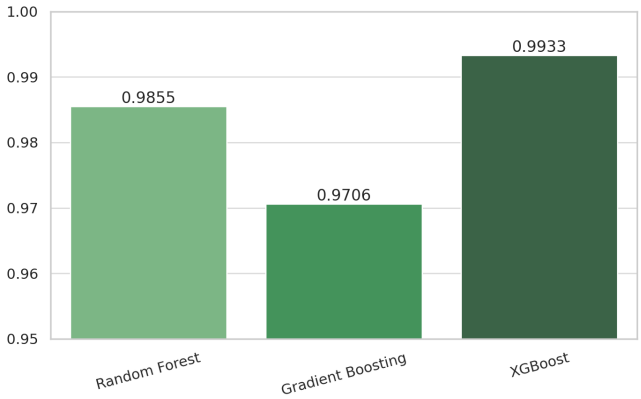

FIGURE 6. Precision Analysis: XGBoost attains the highest precision, reflecting its strong ability to minimize false positives. Random Forest also performs well, with Gradient Boosting trailing slightly.

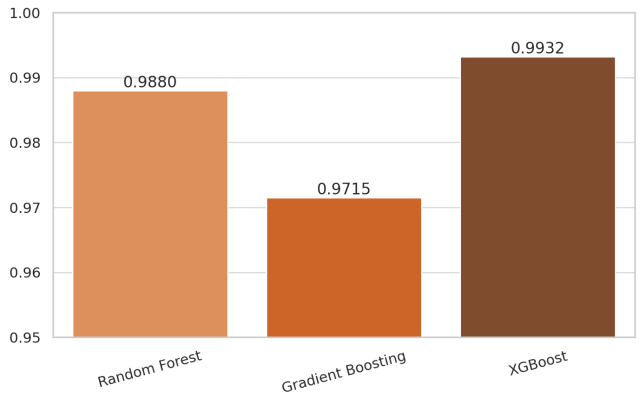

FIGURE 7. Recall Evaluation: All three models achieve high recall scores, with XGBoost capturing nearly all actual positives, indicating its effectiveness in avoiding false negatives.

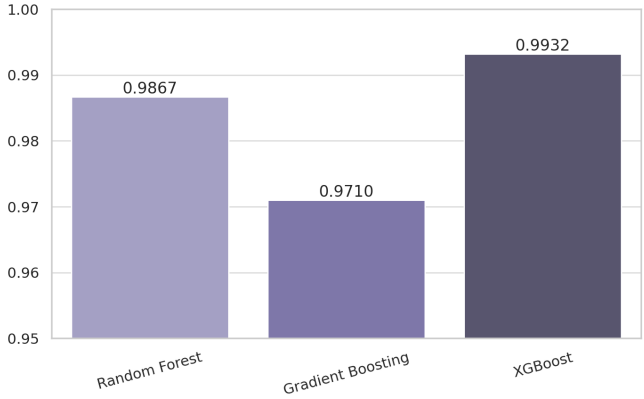

FIGURE 8. F1-Score Comparison: XGBoost shows the highest F1-score, suggesting it offers the best trade-off between precision and recall. Random Forest performs similarly, while Gradient Boosting remains stable but slightly lower.

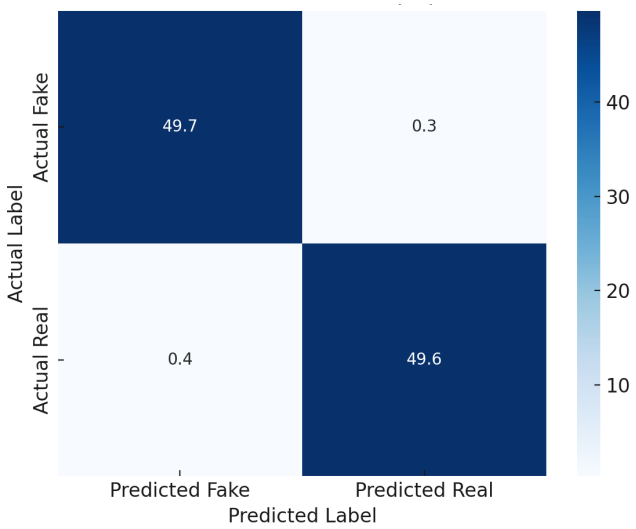

FIGURE 9. Confusion Matrix for XGBoost: The model demonstrates strong performance with high true positive and true negative rates. Only a small number of misclassifications occur, indicating minimal false positives and false negatives.

I. COMPARATIVE RESULTS

To evaluate the effectiveness of the proposed system, we compared its performance with several recent deepfake audio detection models presented in the literature. The comparison is based on the **accuracy** reported by each author using different datasets and methodologies.

TABLE 6. Accuracy Comparison of Deepfake Audio Detection Models

Author	Year	Accuracy (%)
Hamza et al. [1]	2022	98.5
Kilinc et al. [7]	2023	94.2
Pham et al. [5]	2024	97.0
Mcuba et al. [3]	2024	86.9
Qais et al. [6]	2022	72.5
Our Model (XGBoost)	**2025**	**99.3**

The above figure graphically contrasts the accuracy of our XGBoost-based model with the models in the literature. Our

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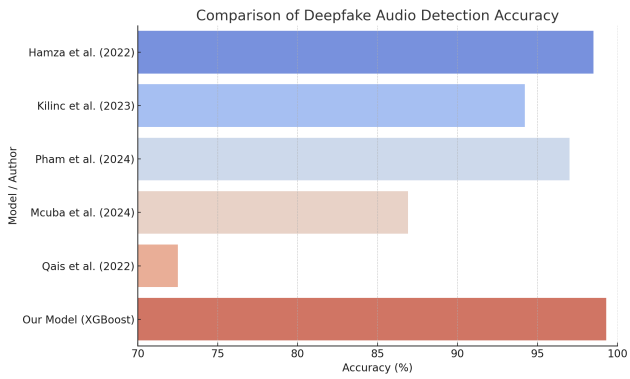

FIGURE 10. Graphical comparison of deepfake audio detection accuracy across multiple studies. Our XGBoost model achieves the highest performance.

model performs better than all the prior methods and offers the best accuracy of **99.3%**. Most importantly, it is more than **0.8%** better than the second best-performing model (Hamza et al., 2022), and hence the detection potential is better.

This improvement in performance is owed to high-resolution Mel-spectrogram features and the effectiveness of ensemble models such as XGBoost, which can learn minute differences between synthetic and real audio. Although competitive results have been achieved by certain deep learning-based methods (e.g., CNN+LSTM), they are usually more computationally intensive and take longer to train. By contrast, our model continues to be lightweight and explainable and thus ideal for real-time and resource-limited deployment environments.

J. DISCUSSION

The experimental results illustrate how deep audio feature integration using techniques like Mel-spectrograms can facilitate standard machine learning models to detect deepfake audio with highly accurate and efficient results. Specifically, the XGBoost model was identified to be robust and consistent in detecting very subtle differences between real and synthetic audio with highly precise accuracy.

The improved detection ability in all of the models also speaks to the quality and diversity of the dataset, which contained mixed, realistic, and varied deepfake examples. Diversity played a critical role in learning generalizable models with the ability to recognize outright as well as covert manipulation. Further, ensemble methods possessed improved decision boundaries and were less prone to overfitting than individual classifiers.

In contrast to computationally expensive and explainable deep learning-based methods, models used in this work provide transparent decision-making, rapid training, and simple deployment. This makes the system proposed here appropriate for real-time application like voice verification, media validation, and forensic examination.

But the system now assumes reasonably clean audio and binary classification (real or fake). Future directions involve background noise robustness, multilingual detection support, and adaptation to changing synthesis techniques, to provide long-term sustainability in increasingly complex audio backgrounds.

A notable limitation of this work is the restricted diversity of the DEEP-VOICE dataset, which includes only eight public figures. This narrow demographic scope may limit the model's generalizability to broader, real-world populations with varying accents, speaking styles, and languages. Future work should incorporate more comprehensive and multilingual datasets to ensure robust detection across diverse audio sources.

VII. ETHICAL IMPLICATIONS AND DUAL-USE CONCERNS

The rise of deepfake detection technology brings along some important ethical questions. While the main aim of this tech is to protect public trust, ensure media authenticity, and enhance digital security, it also presents a tricky dual-use issue. The very algorithms and insights designed for detection could be exploited by bad actors to create even more convincing deepfakes.

Furthermore, when implementing these processes, we must prioritize transparency, accountability, and fairness. If the algorithms aren't trained on a diversity of data, there is a real risk of surveillance misuse, false charges, or biased detection against specific languages, accents, or demographic groups. Consequently, ethical use requires open evaluation criteria, ongoing checks for bias and drift, and transparency about the makeup of the dataset.

It's critical to strike a balance between innovation and appropriate management. We support the development of clear policy frameworks, collaboration between tech experts and ethicists, and public education to ensure that deepfake detection tools promote the truth without endangering privacy or rights.

VIII. LIMITATIONS

Even if the suggested system performs well, there are certain issues that need to be addressed. First off, only eight public persons' audio samples are included in the DEEP-VOICE dataset used in this investigation. This restricted scope reduces the training data's linguistic and demographic diversity, which may have an impact on the model's capacity to generalize to larger populations. Furthermore, the studies were carried out in comparatively controlled settings with minimal artificial audio and background noise. Adapting to noisy surroundings and using more sophisticated deepfake generating techniques may be necessary for real-world deployment. Future research can improve resilience by enlarging the dataset and investigating multilingual, spontaneous speech samples.

IX. CONCLUSION

This study shows that traditional machine learning techniques can detect deepfake audio without the high computa-

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tional cost of deep neural networks. XGBoost performed better than the other models in generalization and stability. This is mainly because it can use the rich patterns found in mel-spectrogram data. However, for the system to work in real-world situations, it must handle challenges like background noise, accent variations, and new synthesis techniques. The current models work well under controlled conditions. To improve resilience, this work should include deep learning architectures, multilingual datasets, and real-time inference pipelines.While the current study uses classical ensemble methods for their interpretability and efficiency, future research should benchmark them against deep learning models such as Wav2Vec 2.0 or AASIST-SSL to evaluate gains in robustness and speaker generalization. More research is needed to keep up with the misuse of generative voice technologies in critical situations.

We also note the absence of direct benchmarking against ASVspoof baselines such as CLAD, AASIST, or SE-ResNet. These architectures represent state-of-the-art in deepfake detection and will be included in future evaluations to assess how our interpretable ensemble models compare against more complex deep learning counterparts.

A. ETHICAL IMPLICATIONS AND DUAL-USE DISCUSSION

The rise of deepfake detection technologies is essential for combating misinformation and safeguarding our online identities. Yet, it also brings up important ethical dilemmas and challenges related to dual-use. On one side, these tools can help maintain media integrity, bolster democratic institutions, and improve voice authentication systems. On the flip side, there's a danger they might be misused for surveillance, censorship, or even to falsely accuse people during tense situations.

There's a genuine concern about initiating an arms race between detection systems and generative models. Scammers could always figure out how to circumvent these detection tools. Additionally, if the systems rely on biased or unrepresentative data, they have the potential to unjustly misflag or miss content from linguistic or cultural groups that are already underrepresented.

It's important that we design and deploy these systems with transparency, fairness, and accountability in mind. We must convene diverse groups of stakeholders—such as tech-savvy folks, policymakers, ethicists, and civil rights leaders—to ensure we're using these tools responsibly. Subsequent iterations of detection models must be accompanied by audit logs, publicly shared benchmarking datasets, and explicit usage licenses to prevent misuse.

X. FUTURE WORK

While here in this project it has been shown how far the machine learning models can be pushed in attempting to detect deepfakes with the assistance of mel spectrograms, there are some things which would need to be added in the future. These involve the utilization of more sophisticated architectures such as Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) that can learn more sophisticated space and time patterns from the audio signals. It must also be scaled up by increasing the size of the dataset as well. Additional audio recordings from other languages, accents, recording settings, and stages of deepfake creation can make the models more robust and generalizable. Pitch-shifting, time-stretching, or adding ambient noise can mimic acoustics and reduce model bias. Real-time deployment and detection is one such upcoming trend. Deploying a trained model into production-level, tiny-sized deployments such as web apps, mobile apps, or REST APIs is one such upcoming trend. This would make real-world deployment possible in voice-based authentication platforms, media verification software, or digital forensics servers. Pruning, quantization, or ONNX/TFLite conversion are some of the techniques available that can be utilized to achieve this. for performance enhancements. Lastly, it can make the model more interpretable by using XAI techniques. Users and stakeholders will have an understanding of what segment of the spectrogram or audio has been used to make a decision to label it because it will build confidence in the system particularly for high-risk use cases such as forensic examination, journalism, and cybersecurity. As deepfake voice goes on improving, should have detection methods and counteracting its effect. Furthermore, future studies should incorporate direct benchmarking against state-of-the-art baselines from the ASVspoof challenge such as CLAD, AASIST, and SE-ResNet. This comparison will help position the proposed lightweight model relative to advanced deep learning architectures in terms of generalization, inference speed, and robustness.

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